

MICROSOFT FABRIC E2E PROJECT GUIDE

Enterprise Pharmaceutical Commercial & Sales Analytics

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1. Background & Architecture

This end-to-end data engineering and analytics solution leverages Microsoft Fabric's Medallion Architecture (Landing Lakehouse → Staging Warehouse → Presentation Warehouse → DirectLake Power BI) to ingest, clean, validate, and report on enterprise-scale pharmaceutical dispensing and prescription data.

2. Ingestion & Lakehouse Setup

- Create workspace: **Pharma_E2E_Analytics_Workspace**
- Create Lakehouse: **ProjectLakehouse**
- Under **Files**, create folder **pharma_sales_records** for monthly Parquet files.
- Under **Files**, create folder **pharma_lookups** and upload **pharma_product_lookup.csv**.

3. Staging Ingestion Pipelines

A. Product Dimension Lookup Pipeline (pl_stg_lookup)

Copy activity: Source = Lakehouse (Files/pharma_lookups/pharma_product_lookup.csv) → Destination = ProjectWarehouse (stg.pharma_product_lookup).

B. Incremental Fact Pipeline (pl_stg_processing_pharma_data)

Ingests monthly files dynamically using expression: `@concat('pharma_sales_', variables('v_date'), '.parquet')` with pre-copy script `delete from stg.pharma_sales_data`.

4. Stored Procedures: Data Hygiene & Metadata

A. Outlier Removal Stored Procedure

```
CREATE PROCEDURE stg.data_cleaning_pharma @start_date DATETIME2, @end_date DATETIME2 AS BEGIN DELETE FROM stg.pharma_sales_data WHERE tpep_pickup_datetime < @start_date OR tpep_pickup_datetime >= @end_date; END;
```

B. Metadata Audit Logging Stored Procedure

```
CREATE PROCEDURE metadata.insert_staging_metadata @pipeline_run_id VARCHAR(255), @table_name VARCHAR(255), @processed_date DATETIME2 AS BEGIN INSERT INTO metadata.processing_log (pipeline_run_id, table_processed, rows_processed, latest_processed_pickup, processed_datetime) SELECT @pipeline_run_id, @table_name, COUNT(*), MAX(tpep_pickup_datetime), @processed_date FROM stg.pharma_sales_data; END;
```

5. Presentation Layer (Dataflow Gen2 / SQL SP)

Create the presentation table in **ProjectWarehouse**:

```
CREATE TABLE dbo.pharma_sales_data ( trx_id VARCHAR(50), vendor VARCHAR(100), dispense_date DATE, therapeutic_area VARCHAR(100), drug_name VARCHAR(150), market_segment VARCHAR(100), payment_method VARCHAR(50), units_dispensed INT, transit_distance FLOAT, net_amount FLOAT );
```

Fast-Path Presentation Load Stored Procedure (Production Alternative):

```
CREATE PROCEDURE dbo.process_presentation_pharma AS BEGIN INSERT INTO dbo.pharma_sales_data SELECT s.trx_id, CASE s.VendorID WHEN 1 THEN 'Manufacturer Direct' WHEN 2 THEN 'McKesson' WHEN 3 THEN 'AmerisourceBergen' WHEN 6 THEN 'Cardinal Health' WHEN 7 THEN 'Specialty Direct' ELSE 'Unknown' END AS vendor, CAST(s.tpep_pickup_datetime AS DATE) AS dispense_date, l.TherapeuticArea AS therapeutic_area, l.DrugName AS drug_name, l.MarketSegment AS market_segment, CASE s.payment_type WHEN 1 THEN 'Commercial Insurance' WHEN 2 THEN 'Medicare Part D' WHEN 3 THEN 'Medicaid' WHEN 4 THEN 'Out-of-Pocket / Cash' WHEN 5 THEN 'Government / VA' ELSE 'Unknown' END AS payment_method, s.passenger_count AS units_dispensed, s.trip_distance AS transit_distance, s.total_amount AS net_amount FROM stg.pharma_sales_data s LEFT JOIN stg.pharma_product_lookup l ON s.ProductID = l.ProductID; END;
```

6. Semantic Model & Executive Power BI Report

Core DAX Measures in pharma_sales_model:

- Total Prescriptions = COUNTROWS('pharma_sales_data')
- Total Net Revenue = SUM('pharma_sales_data'[net_amount])
- Total Units Dispensed = SUM('pharma_sales_data'[units_dispensed])
- Avg Revenue per Rx = DIVIDE([Total Net Revenue], [Total Prescriptions], 0)

Executive Dashboard Layout:

- **Slicers:** Dispense Date (Between), Therapeutic Area (Dropdown/Tiles), Vendor (Tiles), Payment Method (Tiles).
- **KPI Cards:** Total Net Revenue (\$), Total Prescriptions (TRx), Total Units Dispensed, Avg Revenue / Script.
- **Visual 1:** Donut chart displaying Revenue Share by Therapeutic Area.
- **Visual 2:** Multi-line / clustered column chart displaying Daily Revenue Trend by Payer Channel.
- **Visual 3:** Product Performance Matrix with row drill-down (Therapeutic Area → Drug Name → Market Segment).